PyPSA-Libya: An Open-Source Energy System Model for National Decarbonization and Hydrogen Export Pathways

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Abstract

Libya possesses significant renewable energy potential that remains largely untapped. This paper presents PyPSA-Libya, a national-scale model built on the PyPSA-Earth framework, designed to explore optimal transition pathways for the Libyan power sector. We present two primary studies: (1) an electricity-only capacity expansion analysis comparing a zero-emission constraint against an unconstrained baseline at 36-cluster resolution, and (2) a sector-coupled hydrogen export study evaluating the levelized cost of exporting 70 TWh/year of hydrogen to Europe under three technological scenarios. Results indicate that while solar PV dominates the capacity mix due to favorable 2030 cost assumptions, the integration of hydrogen production significantly alters the grid's storage requirements. The model, data, and configuration files are released under an open-source license to support transparent energy planning in Libya.

Keywords: Energy system modelling, Libya, Green Hydrogen, Decarbonization, PyPSA-Earth

1. Introduction

The global imperative to mitigate climate change has accelerated the transition from carbon-intensive fossil fuels to renewable energy sources, in line with the objectives of the Paris Agreement [1, 2]. Petro-state economies face a distinct dual challenge during this transition: they must decarbonize their domestic power sectors while simultaneously diversifying their economies away from oil and gas dependence [1, 3]. Libya represents a critical case study of this paradigm. The country's political economy relies almost entirely on conventional fossil fuels, with the oil sector contributing approximately 95% of export earnings and 65% of Gross Domestic

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Product (GDP) [1]. Over the past decade, political instability and administrative division have severely affected Libya's energy sector. Armed conflicts and unilateral shutdowns of major oil fields (such as those in early 2020 causing financial losses exceeding $9 billion) have repeatedly undermined the reliability of fuel supplies for electricity generation [1].

Concurrently, Libya's domestic power sector, monopolized by the state-owned General Electricity Company of Libya (GECOL), is facing a severe operational crisis [1]. The power grid relies completely on fossil fuels, with natural gas driving approximately 63% of generation and oil-fired plants accounting for 29% [1]. Rapid growth in energy demand, driven by rising temperatures and population growth, has led to a persistent peak-load electricity deficit of approximately 1,200 MW, resulting in frequent and widespread blackouts [1]. Furthermore, Libya's environmental footprint is substantial; in 2018, its per capita CO2 emissions amounted to 8.73 tonnes, surpassing the Chinese national average of 7.95 tonnes [1]. Although the Renewable Energy Authority of Libya (REAoL) set an ambitious target in 2010 to integrate a 10% renewable share (equivalent to 2,219 MW) into the national mix by 2025, this initiative was completely derailed by the civil conflict that erupted in 2011 [1].

Despite these challenges, Libya is exceptionally well-positioned to pivot toward a sustainable energy economy. The country possesses some of the highest solar and wind resource potentials in the world. The average global horizontal irradiation (GHI) over the Libyan territory yields an average potential solar photovoltaic (PV) productivity of 1.9 MWh/kW/year (with Al Kufra in the southeast reaching an optimal annual productivity of 2,100 kWh/kWp), while onshore wind potentials reach up to 500 W/m^2 along its southernmost borders, coastal Darna, and the Western Mountain region [1]. Recognizing these vast resources, pioneering research in the late 1980s first proposed a solar-hydrogen energy system for Libya, emphasizing that converting solar power to hydrogen would not only extend the lifespan of fluid fossil fuel reserves as raw chemical feedstocks but also position Libya as a leading exporter of clean hydrogen to Europe in lieu of oil and gas [3].

To evaluate the modern technical and economic feasibility of this transition, we introduce `PyPSA-Libya`, a high-resolution, open-source energy system model of Libya's power and hydrogen network. Built as a country-specific extension of the global `PyPSA-Earth` framework [4], our model is divided into two primary, complementary studies designed to address both domestic and international planning questions:

1. Electricity-only capacity expansion (Part 1): This study models Libya's power sector in isolation at a high spatial resolution of 36 clustered nodes. It compares a zero-CO2 emission constraint (Case 1) against an unconstrained fossil-heavy baseline (Case 2) under 2030 technology-cost assumptions, capturing transmis-

sion bottlenecks and identifying the optimal mix of solar, wind, and storage needed for complete domestic grid decarbonization.

2. Sector-coupled hydrogen export (Part 2): This study models the spatial and economic implications of exporting 70 TWh/year of renewable or low-carbon hydrogen from Libya to European markets via maritime shipping [4]. Formulated at a 12-cluster resolution to maintain computational tractability, we evaluate three technology scenarios representing strictly green electrolysis (Scenario A), electrolysis coupled with carbon-captured steam methane reforming (SMR CC) representing blue hydrogen (Scenario B), and an unconstrained scenario representing grey, blue, and green pathways (Scenario C) .

To ensure robustness, our model incorporates a `connectivity_upgrade` workflow to resolve topological gaps and isolated buses inherent in OpenStreetMap (OSM) grid data for Libya. We report the Levelized Cost of Electricity (LCOE) and apply a paired-run differencing methodology to isolate the Net Levelized Cost of Hydrogen (Net LCOH) directly attributable to the export obligation. The model code, parameters, and results are released under an open-source license to establish a transparent, reproducible benchmark for national decarbonization and cross-border energy partnerships.

2. Related work

The design of the `PyPSA-Libya` model builds upon three main pillars: (i) localized studies on Libyan renewable and hydrogen potentials, (ii) advancements in open-source spatial-temporal modeling, and (iii) global hydrogen trade pathways and underground storage economics.

2.1. Renewable potential and hydrogen in Libya

Early national energy assessments by Eljrushi and Veziroglu [3] mathematically simulated the dynamic interaction between population growth and fossil fuel depletion, demonstrating that Libya would face severe energy supply deficits without a strategic transition toward solar-hydrogen production. Almaktar and Shaaban [1] modernized these estimates through GIS and meteorological evaluations, categorizing Libya's solar, wind, biomass, and geothermal capacities as essential non-rivalry energy alternatives.

While macro-level studies established these resource potentials, localized techno-economic analyses remain rare. Rahil et al. [5] evaluated off-peak forecourt electrolysis in Darnah under 2015 and 2030 cost horizons; however, their scope was restricted to isolated refueling stations without capturing nationwide transmission

grid dynamics or spatial sector coupling. Our model bridges this gap by representing the full national transmission topology and co-optimizing domestic and export infrastructure.

2.2. Open-source modeling and the PyPSA framework

Proprietary energy planning models (e.g., TIMES, PLEXOS) often present commercial license barriers that limit transparency and collaboration in developing nations. To address this, the Python for Power System Analysis (PyPSA) library was introduced as a fully open-source framework for high-resolution capacity expansion planning. While `PyPSA-Eur` pioneered continental-scale transmission and storage optimization, country-specific forks such as `PyPSA-Spain` [6] demonstrated the necessity of integrating detailed national datasets and nested border-price models to accurately reflect local grid conditions.

Expanding open-source modeling globally, Parzen et al. [7] developed `PyPSA-Earth`, an automated global framework for extracting high-voltage network topologies from OpenStreetMap and mapping spatiotemporal weather potentials. Abdel-Khalek et al. [4] extended this platform into a sector-coupled model incorporating multi-carrier energy vectors (electricity, hydrogen, gas, biomass, oil, and heat) across industry, transport, and residential sectors. Their application establishing Egypt's potential to supply 16% of European hydrogen demand by 2030 forms the methodological benchmark for our sector-coupled export study.

2.3. Global hydrogen trade and geological storage

Cross-border hydrogen trade involves significant techno-economic and infrastructural uncertainties [8]. In the Euro-Mediterranean context, TIAM-ECN modeling by Fattahi et al. [9] identified North Africa as the most cost-effective pipeline supplier of gaseous hydrogen to Europe, with liquefied shipping providing complimentary supply diversification. Alanazi et al. [2] confirmed that international trade could meet over 31% of global decarbonized hydrogen demand by 2050, highlighting North Africa-to-Europe as a key trade corridor. Nevertheless, high conversion and transport losses dictate that hydrogen should be prioritized for hard-to-abate sectors, long-duration storage, and heavy industry rather than direct space heating or light mobility [10].

Managing the volatility of renewable supply under rigid export commitments presents a major operational challenge [11]. Achhammer et al. [12] addressed this by integrating GIS-mapped geological salt caverns into `PyPSA-Earth` for Tunisia, showing that underground hydrogen storage (UHS) reduces levelized export costs by 3.7%, dampens price volatility, and safeguards domestic energy security. By integrating both maritime shipping logistics and underground storage limits, `PyPSA-Libya`

delivers a comprehensive evaluation of how these spatial and geological assets optimize Libya's export resilience.

3. Methodology

The `PyPSA-Libya` model is developed using the open-source `PyPSA-Earth` framework for large-scale energy system optimization. The model minimizes total annualized system costs—comprising capital expenditures (CAPEX) and variable operational costs (OPEX)—across defined temporal and spatial horizons.

3.1. Input data and network topology

The model utilizes OpenStreetMap (OSM) data for high-voltage transmission lines and substations. Preliminary diagnostic testing revealed significant data gaps in the Libyan OSM extract, specifically regarding isolated buses in the southern and desert regions (e.g., Fezzan and Al Kufra). To ensure a fully interconnected national grid backbone, synthetic line connections were added to guarantee that every node retains at least one physical connection to the transmission system.

Renewable resource potentials are derived from the ERA5 reanalysis weather dataset for the year 2013. The spatial resolution varies by study to balance detail with computational tractability:

1. Electricity-only study: 36 clustered nodes, providing high-fidelity insight into internal transmission bottlenecks.
2. Sector-coupled study: 12 clustered nodes, optimized to ensure the complex linear programming (LP) problem fits within 16 GB of memory.

3.2. Techno-economic assumptions

All technology cost assumptions are calibrated to a 2030 planning horizon using the `PyPSA` technology-data repository. We assume a standard Weighted Average Cost of Capital (WACC) of 7.1% (discount rate). The temporal resolution is resampled to 3-hour steps for the electricity study and 6-hour steps for the hydrogen study. This resolution is identified as the "sweet spot" for capturing solar diurnal ramps and battery cycling while preventing the numerical instability observed in 1-hour resolution runs for large-scale matrices.

3.3. Scenario design

The study is divided into two distinct modeling parts:

3.3.1. Part 1: Electricity-only decarbonization

To establish a decarbonization baseline, we compare two cases within the electricity-only framework (no sector coupling):

- Case 1 (Zero-CO2): Imposes a hard `co2_limit` of 0.0, prohibiting all fossil dispatch (OCGT, CCGT, oil) and forcing a 100% renewable system (Solar, Wind, Battery, and H_2 storage).
- Case 2 (Baseline): Imposes a non-binding cap (1.0×10^9 tonnes), allowing the model to freely select the cost-minimal generation mix relying heavily on natural gas.

3.3.2. Part 2: Sector-coupled hydrogen export

Following the methodology of [4], we model the export of 70 TWh/year of hydrogen via maritime shipping. The study is structured into two distinct policy and optimization phases to evaluate how carbon pricing alters technological choices:
Phase 1: Baseline technology scenarios (No CO_2 tax) We first evaluate production economics under existing baseline market structures with zero penalty on carbon venting, isolating the performance of three distinct technological tracks:

- Scenario A (Green H_2): Only water electrolysis is allowed, enforcing a strict annual renewable matching constraint.
- Scenario B (Blue H_2): Force-limits production to electrolysis and Steam Methane Reforming with Carbon Capture (SMR CC) under a rigid CO_2 cap (1.0×10^7 tonnes).
- Scenario C (Grey H_2): An unconstrained baseline pathway allowing electrolysis, SMR CC, and unabated (grey) SMR under a non-binding emissions limit.

Phase 2: Hybrid policy optimization (with CO_2 tax) We then introduce a uniform carbon tax applied directly to all net atmospheric emissions. Rather than running the previous scenarios in isolation, this phase removes all technological boundaries to form a single, integrated hybrid co-optimized scenario. In this unified run, the model is granted access to all three technology pathways simultaneously. The optimizer dynamically co-optimizes investments across green electrolysis, blue SMR CC, and grey SMR, automatically selecting the optimal hybrid technology mix that minimizes the total system cost under the new carbon tax regime.

3.4. Solver configuration

Numerical stability and computational scale are critical challenges in full-year, sector-coupled models. We utilize the HiGHS solver configured with the interior-point method (IPM). For complex runs, we disable the crossover phase and apply relaxed feasibility and optimality tolerances (1×10^{-3}) to prevent numerical stall. The exact linear programming matrix dimensions, variable counts, and computational scaling metrics for Scenarios A, B, and C are detailed in Appendix A.

3.5. Data and cost assumptions

We utilize ERA5 reanalysis weather data for the year 2013, resampled to 3-hour or 6-hour intervals. Technology costs are calibrated to the **2030 horizon**. A critical modification in `PyPSA-Libya` is the `connectivity_upgrade` setting, which ensures that every node in the OSM-derived topology has at least one connection, mitigating the issue of isolated buses common in Libyan data.

4. Results: electricity-only baseline

4.1. Capacity of the energy mix

[Insert discussion here: Case 1 likely shows a massive buildout of Solar PV and Battery storage compared to Case 2's reliance on CCGT/OCGT.]

4.2. Transmission bottlenecks

Utilizing the shadow prices (dual variables) of the transmission constraints, we identify key corridors requiring reinforcement to facilitate 100% renewable penetration.

4.3. System cost breakdown

The overall annualized system cost increases by 31.4% when transitioning from the unconstrained baseline to the carbon-limited scenario, rising from 3.747 to 4.925 billion EUR/year.

As shown in Figure 3, the primary driver behind this cost divergence is the capital expenditure required to firm up renewable generation under strict emission limits. In the Unlimited CO_2 case, fossil-fuel units are heavily dispatched to meet demand, incurring substantial system-wide operational and fuel costs (0.9539 billion EUR/year OPEX) while minimizing the need for energy storage CAPEX (0.001 billion EUR/year).

Conversely, enforcing emission constraints (Limited CO_2) shifts the system from a fuel-intensive operating model to a capital-intensive, solar-dominated paradigm:

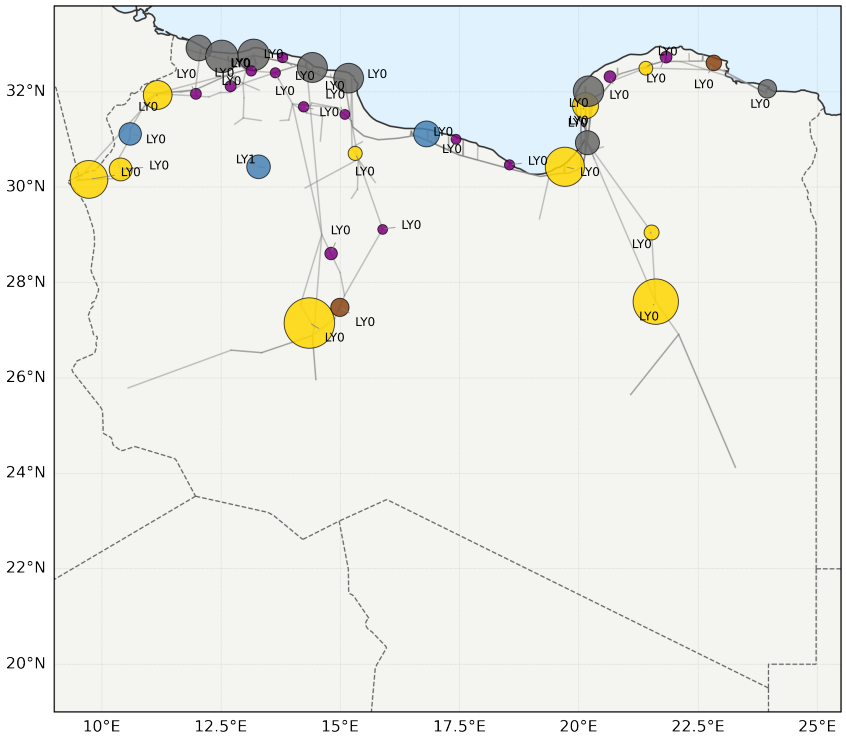


Figure 1: Optimal capacity mix for the 36-cluster electricity-only model (Zero-CO2 Case).

- Solar PV capital expansion: Investment in Solar PV capital expenditure (CAPEX) increases substantially, rising from 0.4211 to 1.4601 billion EUR/year to capture Libya's high solar irradiance.
- Storage integration: To bridge diurnal supply gaps without relying on thermal dispatch, Battery Storage CAPEX surges from near-zero (0.001 billion EUR/year) to 1.3138 billion EUR/year, representing 26.7% of total system expenditure.
- Fossil displacement to backup: While CCGT and oil capital commitments (CAPEX) remain identical across both cases (1.4972 and 0.2518 billion EUR/year, respectively), their operational role switches from baseload/mid-merit dispatch to low-utilization reserve backup. This transition plummets system-wide fuel and variable operating costs (OPEX) to negligible levels (0.0013 billion EUR/year).

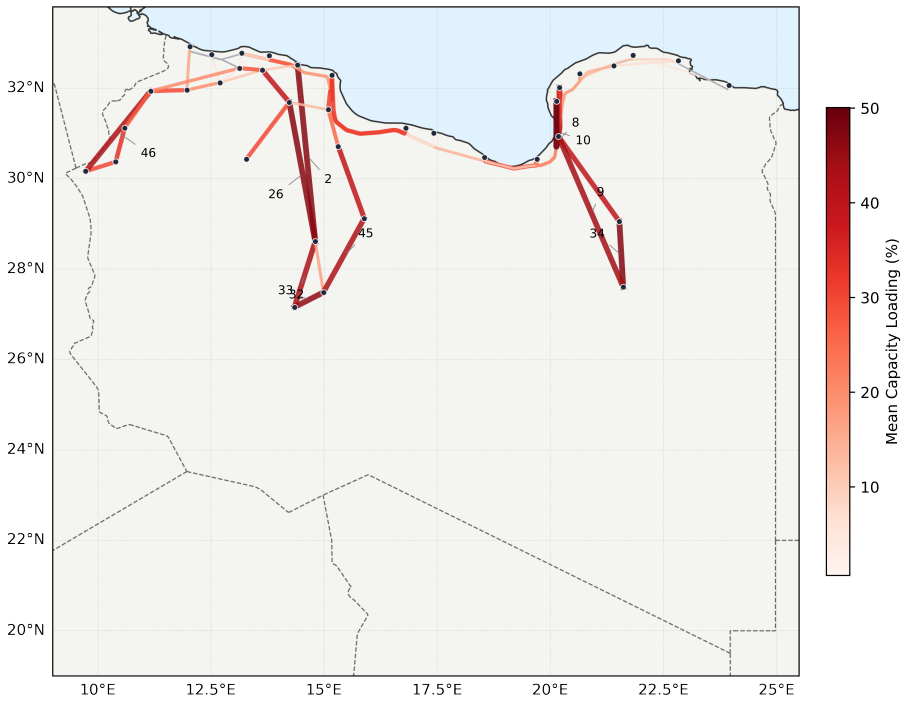


Figure 2: Map of transmission line loading and identified bottlenecks in the Libyan grid.

4.4. Financial analysis

Enforcing zero CO_2 emissions requires a substantial upfront capital commitment to build out renewable energy generation and storage capacity. In contrast, the unconstrained, fossil-heavy system relies on significantly lower initial investment but incurs high ongoing operational and fuel expenses.

To evaluate the economic trade-offs of full grid decarbonization in Libya, a financial feasibility comparison was conducted between the Zero-CO_2 (Limited) scenario and the unlimited baseline (unconstrained CO_2 emissions). Both configurations consider a 25-year operational horizon, a weighted average cost of capital (WACC) of 7.0%, an annual demand of 53.65 TWh, and a benchmark electricity market price of €75.00/MWh. A summary of the financial parameters, annual cash flows, and CO_2 emissions balance is presented in Table 1.

4.5. Economic Takeaways and Policy Implications

The financial and emission findings highlight several key structural insights regarding the energy transition in Libya:

Table 1: Financial parameters, economic performance, and annual CO_2 emissions across the Zero-CO_2 and Unconstrained scenarios.

Metric	Zero-CO_2 (Limited)	Unlimited	Difference (%)
A. Financial Parameters & System Costs			
Overnight CAPEX (€B)[a]	57.494	32.619	+76.3%
Annualized CAPEX (€B/yr)	4.934	2.799	+76.3%
Annual OPEX (€M/yr)	1.330	953.920	-99.9%
Total Annual System Cost (€B/yr)	3.162	1.980	+59.7%
LCOE (€/MWh)	**58.94**	**36.90**	**+59.7%**
B. Profitability & Valuation (at €75.00/MWh Target Tariff)			
Annual Projected Revenue (€B/yr)[b]	4.024	4.024	0.0%
Annual Net Cash Flow (€B/yr)	4.022	3.070	+31.0%
Net Present Value (NPV @ 7%)[c](€B)	-10.617	3.157	-€13.774B
Internal Rate of Return (IRR)	4.86%	8.05%	-3.19% pts
C. Annual CO_2 Emissions Summary			
Total Annual Emissions ($MtCO_2$/yr)	0.000	6.950	-100.0%
System Carbon Intensity (g CO_2/kWh)	0.00	0.13	-100.0%

[a] Overnight CAPEX represents the initial capital expenditure deployed at Year 0.

[b] Projected revenue is calculated using an assumed wholesale electricity price (target tariff) of €75.00/MWh applied to the total annual electricity demand of 53.65 TWh (53.65 TWh).

[c] NPV is calculated over a 25-year project horizon ($T = 25$) at a 7.0% discount rate. It represents the single cumulative present value of all cash flows evaluated at Year 0: NPV $= -\text{CAPEX} + \sum_{t=1}^{25} \frac{\text{Revenue}-\text{OPEX}}{(1+r)^t}$, where Year 0 reflects upfront capital outlay and Years 1–25 reflect discounted operational net cash flows.

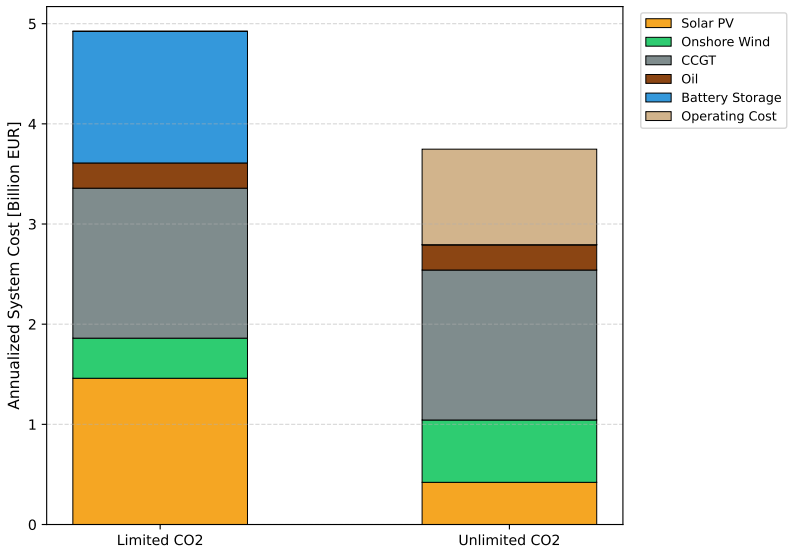


Figure 3: Annualized system cost breakdown by technology for the Limited CO_2 and Unlimited CO_2 electricity-only scenarios.

- Front-loaded CAPEX vs. OPEX trade-off: Achieving a zero-emission grid inflates overnight capital expenditures by 76.3% (€57.494 B vs. €32.619 B) due to extensive solar PV, wind, and battery/H_2 storage investments. However, annual operational expenditures drop by 99.9% (€1.33 M vs. €953.92 M) as fossil fuel purchasing is virtually eliminated.

- The "Green premium" and tariff discrepancy: At the baseline electricity market price of €75.00/MWh, the unconstrained fossil system yields an LCOE of €36.90/MWh (resulting in a positive NPV of €3.157 B and an Internal Rate of Return (IRR) of 8.05%), whereas full decarbonization drives the LCOE to €58.94/MWh (yielding an IRR of 4.86%). This establishes a green premium cost of €22.04/MWh (+59.7%).

- Policy and financing requirements: Although the Zero-CO_2 grid generates higher net annual operational cash flow (€4.022 B vs. €3.070 B), it fails to fully recover its massive initial investment under commercial financing (7% WACC). To make zero-carbon infrastructure bankable (NPV $\geq$ 0), targeted policy interventions are required—such as increasing wholesale power purchase agreements (PPAs) above €59.00/MWh, implementing carbon taxes on fossil fuels, or accessing concessional development financing with interest rates below 4.86%.

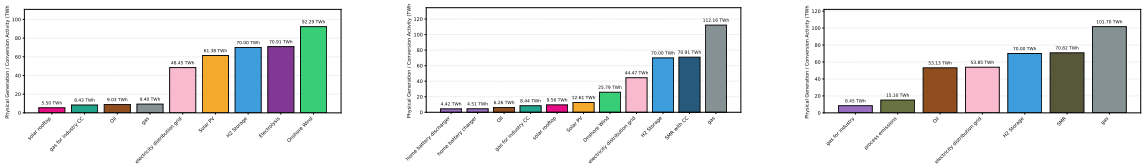
5. Results: Sector-coupled hydrogen export

This section presents the results for a 70 TWh hydrogen export target across Scenarios A, B, and C, highlighting how alternative system configurations influence generation capacity, spatial investment patterns, and economic and environmental performance.

5.1. Energy mix by carrier

The cross-scenario comparison highlights how power supply dispatch transitions across technology carriers under varying system constraints.

Figure 4 illustrates the generation mix by carrier across Scenarios A, B, and C, highlighting the differences in capacity deployment under the respective system constraints.



(a) Scenario A (b) Scenario B (c) Scenario C

Figure 4: Capacity of generation mix by carrier across Scenarios A, B, and C.

5.2. Spatial capacity expansion allocation

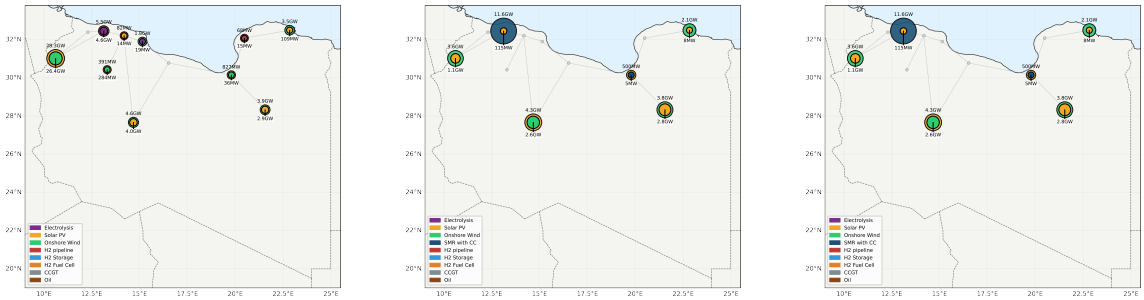
The optimizer allocates hydrogen production near high-irradiance zones and coastal export hubs.

Figure 5 shows the spatial allocation of hydrogen production capacity and supporting infrastructure across the three scenarios, highlighting the concentration of investments near high-irradiance areas and coastal export hubs.

5.3. System cost breakdown

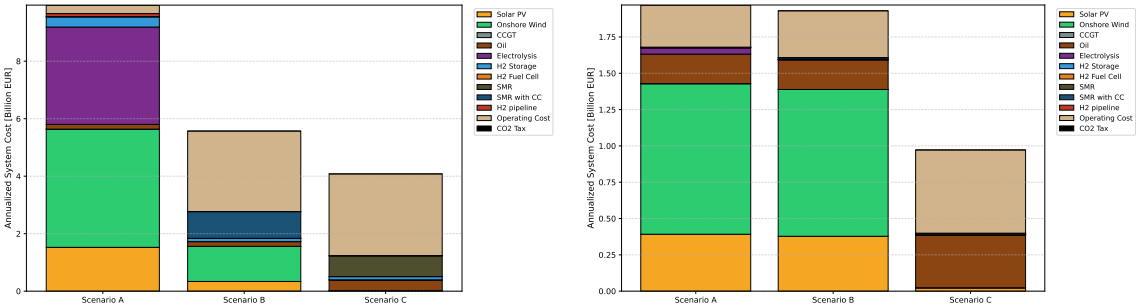
The total system expenditure exhibits notable variations across the evaluated scenarios, driven by the structural allocation of capital assets and operational expenses. To isolate the financial impact of large-scale hydrogen integration, annual expenditures are evaluated under both active export conditions and zero-export baseline conditions.

Under the 70 TWh export demand (Fig. 6a), total annual system expenditures range from 4.080 to 9.952 billion €/yr. Scenario A is heavily CAPEX-driven, dominated by dedicated renewable buildout (4.112 billion €/yr Onshore Wind, 1.525 billion €/yr Solar PV) and electrolysis infrastructure (3.386 billion €/yr). In contrast,



(a) Scenario A (b) Scenario B (c) Scenario C

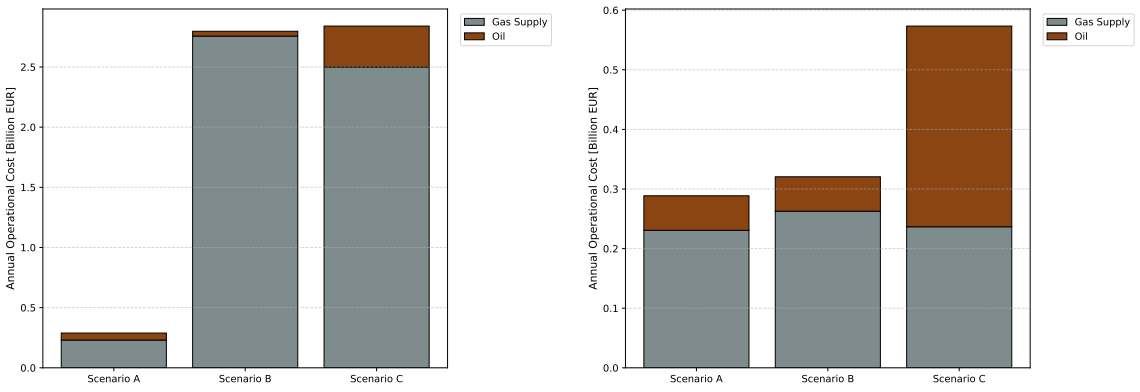
Figure 5: Spatial capacity allocation for hydrogen production and supporting infrastructure.



(a) With hydrogen export (70 TWh) (b) Zero-export baseline case

Figure 6: Total annual system cost comparison across the evaluated scenarios: (a) with hydrogen export at 70 TWh and (b) zero-export baseline case.

Scenarios B and C shift financial burden toward operational expenditures (2.800 and 2.841 billion €/yr, respectively), substituting massive wind/PV investments with grid-supplied electricity and Steam Methane Reforming with Carbon Capture (0.928 billion €/yr in Scenario B). Conversely, the zero-export baseline (Fig. 6b) exhibits substantially lower overall system costs (0.972 to 1.970 billion €/yr), where expenditures across all scenarios remain primarily tied to satisfying domestic electricity demand via modest renewable capacity, legacy oil generation, and minimal operational overhead.



(a) With hydrogen export (70 TWh)

(b) Zero-export baseline case

Figure 7: Operational Expenditure (OPEX) breakdown across the evaluated scenarios: (a) with hydrogen export at 70 TWh and (b) zero-export baseline case.

The OPEX breakdown (Fig. 7) highlights a stark divergence between export and domestic-only cases. Under 70 TWh export demand (Fig. 7a), total OPEX surges from 0.289 billion €/yr in Scenario A to nearly 2.84 billion €/yr in Scenarios B and C, overwhelmingly driven by natural gas costs for SMR and thermal generation. Conversely, without exports (Fig. 7b), system-wide OPEX remains below 0.60 billion €/yr across all scenarios, driven predominantly by baseline legacy fuel demand. Notably, across all cases, primary fuel supply (gas and oil) overwhelmingly dominates operational expenditure, rendering non-fuel OPEX components negligible.

5.4. Levelized cost of hydrogen (LCOH)

To evaluate the economic viability of hydrogen production for export, we compute the net levelized cost of hydrogen across the considered scenarios. This metric provides a standardized basis for comparing production economics under varying grid integration strategies.

To isolate the incremental cost of hydrogen production from existing domestic power grid requirements, we utilize a paired-run subtraction method. For each scenario, the model is solved under the 70 TWh export demand (Fig. 6a) and compared against a corresponding zero-export baseline (Fig. 6b). The Net Levelized Cost of Hydrogen (Net LCOH) is defined as:

$$\text{Net LCOH} = \frac{\text{Total Cost}_{\text{Export}} - \text{Total Cost}_{\text{Baseline}}}{\text{Exported Hydrogen Energy}} \tag{1}$$

This approach ensures that domestic electricity consumers are not modeled as subsidizing the export infrastructure, providing a true marginal cost of hydrogen for the European market.

Table 2: Summary of H_2 Export Costs and Net LCOH Breakdown

Metric	Scenario A	Scenario B	Scenario C
Export Run Cost [Billion €]	9.952	5.571	4.080
Baseline Zero-Export Cost [Billion €]	1.970	1.930	0.972
Net H_2 Cost [Billion €]	7.982	3.640	3.107
Net LCOH [€/MWh]	114.02	52.00	44.39
Net LCOH LHV [€/kg]	3.80	1.73	1.48
Net LCOH HHV [€/kg]	4.49	2.05	1.75

Note: Net H_2 Cost = Export System Cost – Baseline Zero-Export Cost; Net LCOH = Net H_2 Cost / Exported H_2 Energy (70 TWh). Conversions assume H_2 Lower Heating Value (LHV) = 0.03333 MWh/kg and Higher Heating Value (HHV) = 0.03941 MWh/kg.

As summarized in Table 2, the Net LCOH exhibits significant divergence across the evaluated configurations. Across the 70 TWh export target, Scenario A yields the highest Net LCOH at 114.02 €/MWh (3.80 €/kg LHV), driven by the requirement for full renewable off-grid system expansion and dedicated asset over-provisioning. In contrast, Scenario B substantially reduces this metric to 52 €/MWh (1.73 €/kg LHV) through grid interaction and shared flexibility. Scenario C achieves the lowest overall production cost at 44.39 €/MWh (1.48 €/kg LHV), highlighting the strong economic advantage of co-locating generation and leveraging existing national transmission infrastructure.

5.5. Net present value (NPV)

Table 3 presents the financial performance under a dynamic margin framework, applying a constant revenue premium of +€1.00/kg above the levelized cost of hydro-

gen (LCOH LHV) for each scenario over the 21-year operational horizon (2030–2050 at $r = 7.10\%$).

Table 3: Annual operational cash flows, project return metrics, and multi-year discounted financial evaluation by scenario (2030–2050) under a dynamic cost-plus revenue margin pricing model (+€1.00/kg over scenario-specific LCOH at LHV) and an assumed zero CO_2 emission tax, evaluated at a discount rate of $r = 7.10\%$.

Financial Metric	Scenario A	Scenario B	Scenario C
LCOH LHV [EUR/kg]	3.80	1.73	1.48
Selling Price [EUR/kg]	4.80	2.73	2.48
Annual Cost [B EUR/yr]	9.95	5.57	4.08
Annual Revenue [B EUR/yr]	10.08	5.74	5.21
Annual Net Cash [B EUR/yr]	+0.13	+0.17	+1.13
Disct. Cost [B EUR]	106.97	59.88	43.85
Disct. Revenue [B EUR]	108.37	61.71	55.98
Total NPV [B EUR]	+1.40	+1.83	+12.13
Project IRR (%)	10.23%	25.91%	32.50%
Payback Period [Years]	8.51	3.83	3.07

Under this pricing strategy, all three configurations yield positive total system net present values (NPV), but Scenario C demonstrates clear economic superiority. Achieving the lowest levelized cost of hydrogen at lower heating value (LCOH LHV) of €1.48/kg and a selling price of €2.48/kg, Scenario C minimizes annual operational expenditure to €4.08 billion/yr and delivers the lowest discounted system cost of €43.85 billion. This leads to a substantial total system NPV of +€12.13 billion. Furthermore, Scenario C achieves an outstanding Project IRR of 32.50% and rapid capital recovery with a payback period of just 3.07 years, significantly outperforming Scenario B (25.91% IRR; 3.83-year payback) and Scenario A (10.23% IRR; 8.51-year payback).

By contrast, Scenario B (selling price of €2.73/kg) yields an NPV of +€1.83 billion, while Scenario A requires an elevated market price of €4.80/kg to achieve a positive NPV of +€1.40 billion due to its high discounted system cost (€106.97 billion). Consequently, Scenario C remains the most financially resilient option, generating the strongest returns while keeping market-clearing hydrogen export prices competitive.

5.6. Environmental evaluation

In this section, the performance of the model is evaluated in terms of the environmental factors.

5.6.1. CO_2 emissions

As detailed in Table 4, gross CO_2 generation is partitioned into net atmospheric emissions (vented) and geologically sequestered emissions (captured via CCS). Scenarios A and B operate well within the carbon cap ceiling, resulting in identical net atmospheric emissions of 6.82 million tonnes (Mt) while relying on CCUS infrastructure to manage total system generation (capturing 25.25 Mt and 43.56 Mt, respectively). Conversely, the unconstrained baseline in Scenario C yields a significantly higher total gross generation of 135.07 Mt, consisting of 59.15 Mt in atmospheric venting and 75.91 Mt in carbon capture.

Table 4: Gross CO_2 generation partitioned into net atmospheric emissions (vented) and geologically sequestered emissions (captured via CCS), reported in million tonnes (Mt).

CO_2 Accounting Metric	Scenario A	Scenario B	Scenario C
Net Atmospheric CO_2 Emissions (Vented)	6.82 Mt	6.82 Mt	59.15 Mt
Geologically Sequestered CO_2 (Captured)	25.25 Mt	43.56 Mt	75.91 Mt
Total Gross CO_2 Generated (Vented + Stored)	32.07 Mt	50.39 Mt	135.07 Mt

5.6.2. Water consumption

A critical factor in the feasibility of large-scale hydrogen production in arid regions like Libya is the required water supply. In this study, the costs and energy overhead associated with Reverse Osmosis (RO) desalination are integrated directly into the hydrogen supply chain assumptions, accounting for the technology-specific water intensities of electrolysis, SMR with carbon capture, and conventional SMR.

Our analysis indicates that across all three scenarios, total hydrogen output remains virtually identical at approximately 2.128 Mt (70.91–70.92 TWh_{H2}). However, water demand varies significantly based on the technology mix deployed, requiring 21.27 Mm^3 in Scenario A, 15.96 Mm^3 in Scenario B, and 10.65 Mm^3 in Scenario C (Table 5).

Despite the scale of these volumes, the economic impact of water supply is remarkably low. The estimated desalination costs contribute between 0.321% and 0.430% to the total annualized system cost, representing a marginal addition of only €0.0075–0.0150 per kg of hydrogen.

Table 5: Estimated water demand and desalination costs across scenarios.

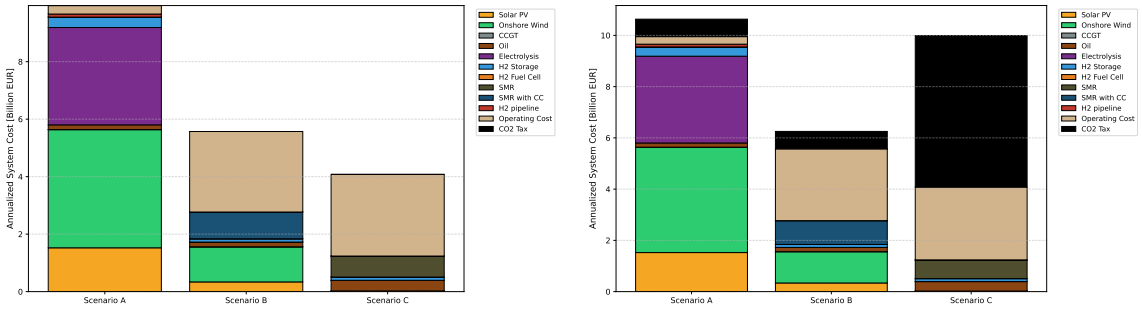
Metric	Scenario A	Scenario B	Scenario C
H_2 Produced (TWh_{H2})	70.91	70.91	70.92
H_2 Mass (Million Tonnes)	2.127	2.128	2.128
Water Demand (Million m^3)	21.27	15.96	10.65
Est. Desalination Cost (M€/yr)	31.91	23.94	15.97

From an energy perspective, the overhead required for RO desalination results in a specific energy consumption of just 0.0175–0.0350 kWh/kg_{H2}. When evaluated against total system electricity demand, desalination accounts for merely 0.033%–0.067%. These findings confirm that while water sourcing requires geographical planning (e.g., coastal co-location of RO plants), water consumption does not act as an economic or energetic bottleneck for Libya's hydrogen supply potential.

5.7. System cost dynamics under CO_2 taxation

The inclusion of a 100 €/tonne CO_2 tax fundamentally alters system cost dynamics, shifting economic penalties away from capital expenditure and toward carbon-intensive operation.

The impact of carbon pricing on total system costs across export scenarios is illustrated in Figure 8.



(a) Without carbon tax

(b) With 100 €/tonne CO_2 tax

Figure 8: Annualized total system cost breakdown across hydrogen export scenarios (70 TWh) under (a) untaxed conditions and (b) a 100 €/tonne CO_2 carbon tax.

- Tax Liability vs. CAPEX Expansion: Unabated fossil reliance drives prohibitive penalty costs. In Scenario C, the CO_2 tax alone reaches 5.915 Billion €/year,

constituting over 59% of its total annual expenditure (9.995 Billion €/year). Conversely, higher capital investments (CAPEX) in renewables and electrolysis (Scenario A: 1.525 Billion €/year Solar PV, 4.112 Billion €/year Wind, 3.386 Billion €/year Electrolysis) limit carbon tax liabilities to just 0.682 Billion €/year.

- Technology Competitiveness: The 100 €/tonne threshold erodes the viability of unabated Steam Methane Reforming (SMR). Scenario B demonstrates that integrating Carbon Capture (SMR-CC) yields the lowest overall system cost (6.253 Billion €/year), balancing modest capital investment (0.928 Billion €/year) with a heavily reduced carbon tax penalty (0.682 Billion €/year).

- Expenditure Composition: Under carbon pricing, high-carbon pathways (Scenario C) transition to an operational-liability model dominated by tax payments, whereas decarbonized options shift system value toward fixed, asset-driven capital expenditure.

5.8. Financial viability under carbon taxation

A comparative summary of the core financial metrics under untaxed conditions (0 EUR/tCO_2) versus a 100 EUR/tonne CO_2 tax regime is presented in Table 6.

Table 6: Comparison of key financial performance metrics across scenarios under zero tax versus a 100 EUR/tonne CO_2 tax ($r = 7.10\%$, 2030–2050 horizon, revenue margin +1 EUR/kg).

	Scenario A		Scenario B		Scenario C	
Financial Metric	No Tax	Taxed	No Tax	Taxed	No Tax	Taxed
Annual System Cost [B EUR/yr]	9.95	10.63	5.57	6.25	4.08	9.99
Total System NPV [B EUR]	+1.40	-5.94	+1.83	-5.51	+12.13	-51.46
Project IRR (%)	10.23%	9.25%	25.91%	20.22%	32.50%	N/A
Payback Period [Years]	8.51	9.13	3.83	4.84	3.07	N/A

N/A indicates that net operating cash flows are strictly negative, resulting in no breakeven point; thus, IRR and Payback Period are mathematically invalid to calculate.

The introduction of carbon pricing severely undermines project bankability across all pathways, though to vastly different degrees:

- Catastrophic Impact on Fossil Pathways: Scenario C experiences total economic collapse under carbon pricing. The 5.915 Billion EUR/year tax liability drives total annual system costs up to 9.99 Billion EUR/year, rendering the

total system NPV deeply negative (−51.46 Billion EUR). Because annual operating cash flows remain strictly negative (−3.41 Billion EUR/year), the system never reaches a financial breakeven point, yielding non-computable (N/A) figures for IRR and Payback Period.

- Resilience of Decarbonized Pathways: Scenarios A and B absorb the tax policy far more effectively due to minimal residual emissions. While both transition to negative total system NPVs (−5.94 B EUR and −5.51 B EUR, respectively), they maintain positive net operating cash flows. Scenario B retains the most robust financial profile with a 20.22% Project IRR and a rapid payback period of 4.84 years.

6. Results: Hybrid pathway for hydrogen export

This section examines the results obtained under the hybrid scenario, which combines a 70 TWh hydrogen export target with a 100 €/tonne CO_2 carbon tax. The analysis focuses on the resulting generation capacity, spatial capacity allocation, and system costs under this combined policy and export setting.

6.1. Energy mix by carrier

The cross-scenario comparison highlights how power supply dispatch transitions across technology carriers under varying system constraints.

Figure 9 presents the generation capacity mix by carrier under the hybrid scenario, illustrating the contribution of different technology carriers to the overall energy supply.

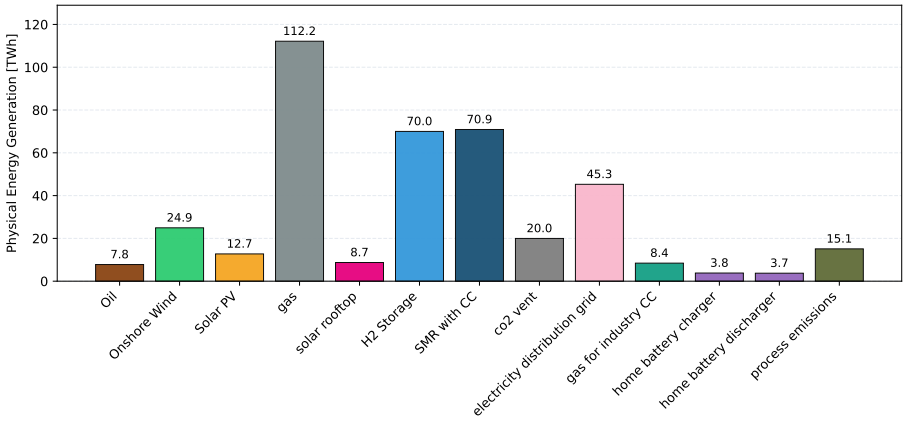


Figure 9: Capacity of generation mix by carrier under hybrid scenario

6.2. Spatial capacity expansion

The optimizer allocates hydrogen production near high-irradiance zones and coastal export hubs.

Figure 10 depicts the geographical distribution of hydrogen production capacity and associated infrastructure under the hybrid scenario, revealing the spatial pattern of investment across the system.

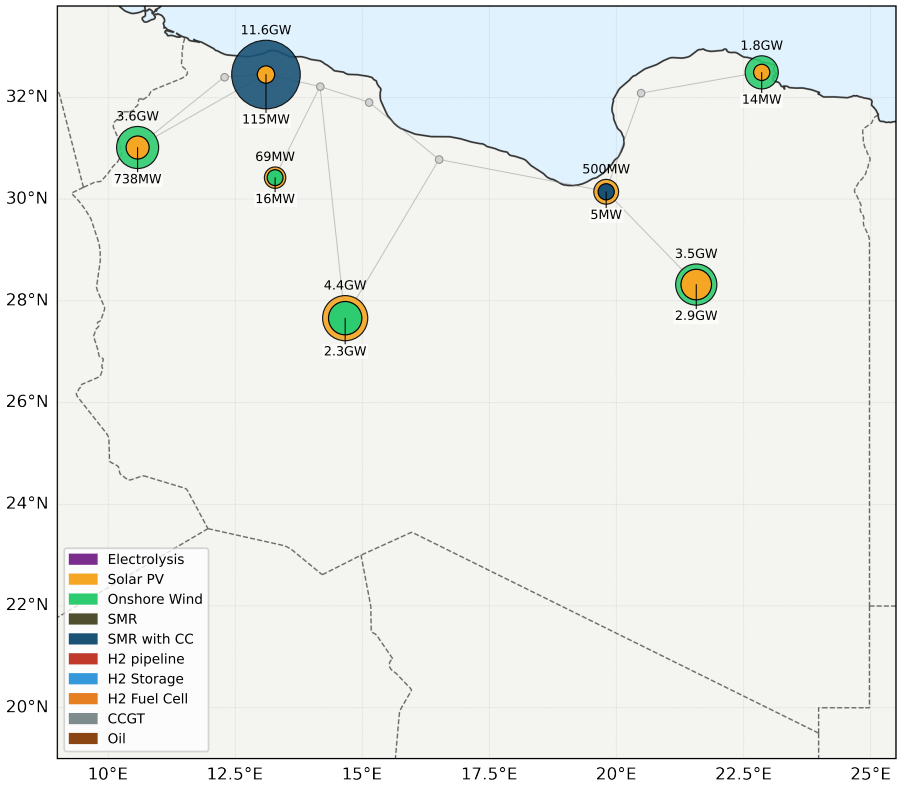
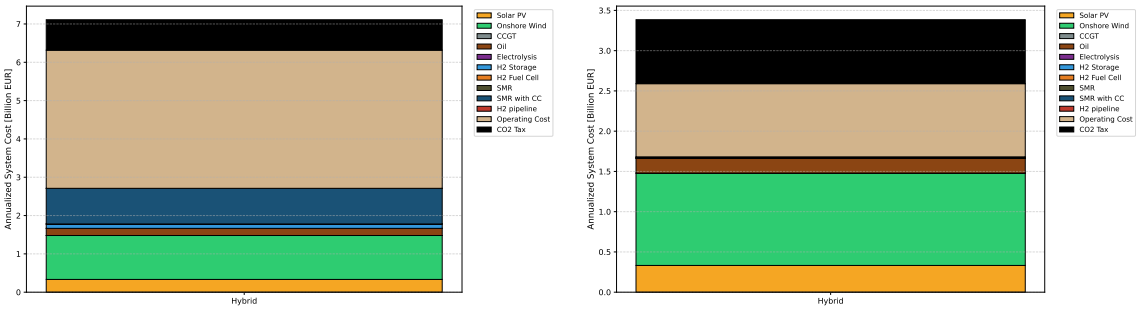


Figure 10: Spatial capacity allocation for hydrogen production and supporting infrastructure under the hybrid scenario

6.3. System cost breakdown

Integrating a combined policy framework—pairing a 70 TWh hydrogen export mandate with a 100 €/tonne CO_2 carbon tax—reshapes the overall system economics by striking a balance between capital expenditure (CAPEX) investments and operational emission penalties.

The annualized system cost structure for the hybrid scenario is shown in Figure 11, comparing the 70 TWh export case against the zero-export baseline.



(a) With hydrogen export (70 TWh)

(b) Zero-export baseline case

Figure 11: Annualized total system cost breakdown under the hybrid scenario (100 €/tonne CO_2 carbon tax): (a) with 70 TWh hydrogen export and (b) zero-export baseline case.

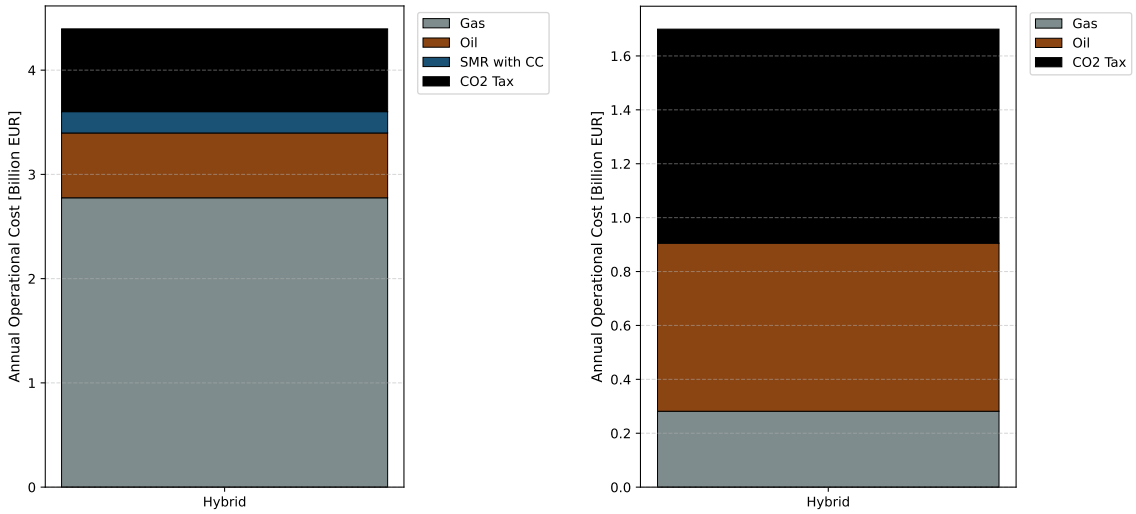
- Export Burden and Scale Expansion: Meeting the 70 TWh external hydrogen demand elevates the total annual system cost to 7.110 Billion €/year, representing a 110.1% increase over the zero-export baseline (3.384 Billion €/year). This expenditure surge is overwhelmingly driven by thermal conversion and operational scaling rather than pure renewable expansion.

- SMR with Carbon Capture Dominance: To satisfy the export volume under strict carbon penalties without over-investing in dedicated renewable generation, the optimizer relies heavily on Steam Methane Reforming with Carbon Capture (SMR-CC). Capital allocation for SMR-CC stands as the largest single technology investment at 0.928 Billion €/year, whereas green electrolysis CAPEX remains minimal (0.0003 Billion €/year).

- Operational Cost and Tax Liabilities: Operating costs constitute the single largest cost component in the export case at 3.6045 Billion €/year (50.7% of total system expenditure), reflecting fuel inputs for fossil-based pathways. Meanwhile, the 100 €/tonne carbon tax liability settles at 0.7943 Billion €/year in both export and non-export cases, demonstrating that carbon capture deployment successfully caps carbon penalties despite the massively scaled supply output.

- Renewable Integration Profile: Onshore wind forms the backbone of electricity CAPEX (1.1461 Billion €/year under export vs. 1.1464 Billion €/year baseline), outscaling solar PV investment (0.3333 Billion €/year export vs. 0.3324 Billion €/year baseline) across both operational regimes.

The hybrid OPEX breakdown (Fig. 12) illustrates that 70 TWh export demand



(a) With hydrogen export (70 TWh)

(b) Zero-export baseline case

Figure 12: Operational Expenditure (OPEX) breakdown for the hybrid model (100 €/tonne CO_2 carbon tax): (a) with 70 TWh hydrogen export and (b) zero-export baseline case.

surges operational expenses from 1.699 to 4.396 billion €/yr, heavily dominated by natural gas input for SMR with CC (2.776 billion €/yr), alongside constant oil costs (0.622 billion €/yr) and fixed CO_2 tax liabilities (0.794 billion €/yr). Notably, primary fuel and carbon penalties overwhelmingly account for the system's operational expenditure, rendering non-fuel OPEX components negligible across both export regimes.

6.4. Levelized cost of hydrogen (LCOH) under hybrid pathway

To evaluate the economic viability of hydrogen production for export under combined carbon pricing and export constraints, we compute the net levelized cost of hydrogen for the hybrid scenario. This metric provides a standardized basis for evaluating export competitiveness under a 100 €/tonne CO_2 carbon tax regime.

To isolate the incremental cost of hydrogen production from existing domestic power grid requirements, we utilize a paired-run subtraction method based on Eq. 1. The model is solved under the 70 TWh export demand and compared against a corresponding zero-export baseline.

This subtraction approach ensures that domestic electricity consumers are not modeled as subsidizing the export infrastructure, providing a true marginal production cost for the European market.

Remark: The LCOH calculations presented here reflect purely capital and operational production costs and explicitly exclude CO_2 carbon tax liabilities.

Table 7: Summary of H_2 Export costs and net LCOH breakdown for the hybrid pathway

Metric	Hybrid Scenario
Export Run Cost [Billion €]	6.316
Baseline Zero-Export Cost [Billion €]	2.590
Net H_2 Cost [Billion €]	3.726
Exported H_2 Energy [TWh]	70.0
Net LCOH [€/MWh]	53.22
Net LCOH LHV [€/kg]	1.77
Net LCOH HHV [€/kg]	2.10

Note: Net H_2 Cost = Export System Cost – Baseline Zero-Export Cost; Net LCOH = Net H_2 Cost / Exported H_2 Energy. LCOH figures exclude CO_2 carbon tax liabilities. Conversions assume H_2 Lower Heating Value (LHV) = 0.03333 MWh/kg and Higher Heating Value (HHV) = 0.03941 MWh/kg.

As summarized in Table 7, the hybrid policy environment results in a Net LCOH of 53.22 €/MWh, which translates to 1.77 €/kg on a Lower Heating Value (LHV) basis and 2.10 €/kg on a Higher Heating Value (HHV) basis. Subtracting the baseline domestic grid cost (2.590 Billion €) from the full export system expenditure (6.316 Billion €) yields an incremental export system cost of 3.726 Billion €.

This unit production cost highlights the economic trade-off of the hybrid regime: while the 100 €/tonne CO_2 tax restricts unabated fossil options and forces investment in carbon capture infrastructure (SMR-CC), the ability to share grid flexibility prevents the extreme capital premiums associated with purely off-grid renewable pathways. Consequently, the resulting price point of 1.77 €/kg LHV positions Libyan hydrogen exports competitively against projected European import cost benchmarks.

6.5. Net present value (NPV) under hybrid pathway

Table 8 presents the financial performance metrics for the hybrid scenario under a dynamic cost-plus pricing framework. A constant revenue margin of +1.00 €/kg above the scenario LCOH LHV is applied over the 21-year operational horizon (2030–2050) using a 7.10% real discount rate. While the base LCOH metric reflects unit production costs exclusive of carbon penalties, carbon tax liabilities are separately integrated into the broader system-level cash flow, NPV, IRR, and payback evaluations to ensure comprehensive financial accounting.

Table 8: Annual operational cash flows, project return metrics, and multi-year discounted financial evaluation (2030–2050) for the hybrid scenario under a +1.00 €/kg revenue margin pricing model and a 100 €/tonne CO_2 carbon tax ($r = 7.10\%$).

Financial Metric	Hybrid Scenario
LCOH LHV Cost [€/kg]	1.77
Selling Price (+1 €/kg) [€/kg]	2.77
CO_2 Tax Liability [B €/yr]	0.79
Annual Total System Cost [B €/yr]	7.11
Annual Revenue [B €/yr]	5.83
Annual System Net Cash Flow [B €/yr]	-1.28
Annual Project Net Cash Inflow [B €/yr]	+1.81
PV System Cost [B €]	76.42
PV Gross Revenue [B €]	62.62
Total System NPV [B €]	-13.80
Project IRR (%)	20.69%
Payback Period [Years]	4.74

Note: Revenue margin is added directly to LCOH LHV. System annual costs and subsequent cash flows explicitly include the 100 €/tonne CO_2 carbon tax liability.

Under the combined policy setting, the base LCOH LHV of 1.77 €/kg establishes a dynamic market selling price of 2.77 €/kg. At this price level, annual gross revenues reach 5.83 Billion €/year. When evaluated against the total annual system expenditure of 7.11 Billion €/year (which includes domestic grid obligations and 0.79 Billion €/year in carbon tax liabilities), total system-level net cash flows settle at −1.28 Billion €/year, yielding a multi-year discounted system NPV of −13.80 Billion € (62.62 Billion € PV revenue vs. 76.42 Billion € PV system cost).

However, when evaluating the commercial viability of the dedicated hydrogen export infrastructure, the +1.00 €/kg margin over net production costs generates a strong dedicated annual net cash inflow of +1.81 Billion €/year. Assessing this project-level cash flow stream against net initial capital outlays yields a Project Internal Rate of Return (IRR) of 20.69% and a rapid payback period of 4.74 years.

These results confirm that even under a penalizing 100 €/tonne carbon tax regime, the hybrid export project remains highly bankable under a standard cost-plus commercial off-take agreement. The +1.00 €/kg premium provides more than sufficient financial cushion to secure strong double-digit project returns well above the 7.10% hurdle rate.

6.6. Environmental and resource evaluation under hybrid pathway

This section evaluates the environmental impact and resource demand of the hybrid scenario, focusing on CO_2 emissions balance, carbon capture effectiveness, and fresh water requirements for hydrogen production.

Under the hybrid regime, gross system CO_2 generation is partitioned into net atmospheric emissions (vented) and geologically sequestered emissions (captured via CCS). As detailed in Table 9, total gross generation reaches 52.62 Mt, of which 44.68 Mt (84.9%) is successfully captured and sequestered. The remaining 7.94 Mt of net atmospheric emissions consists primarily of unabated oil generation (5.72 Mt), uncaptured process emissions from SMR-CC (2.03 Mt), and residual industrial gas usage (0.19 Mt). This high capture rate highlights how the 100 €/tonne carbon tax effectively incentivizes deep carbon abatement across the production chain.

Table 9: CO_2 mass balance and vented emissions breakdown for the hybrid scenario.

CO_2 Accounting Metric / Source	Value
Net Atmospheric CO_2 Emissions (Vented)	7.94 Mt
Geologically Sequestered CO_2 (Captured)	44.68 Mt
Total Gross CO_2 Generated (Vented + Stored)	52.62 Mt
Breakdown of Vented Emissions by Source:	
Oil	5.72 Mt
SMR-CC	2.03 Mt
Industrial Gas	0.19 Mt

A critical consideration for hydrogen production in Libya's arid environment is the associated water consumption. Reverse Osmosis (RO) desalination energy overheads and costs are integrated directly into the system model, applying technology-specific water intensities (10 L/kg for electrolysis, 7.5 L/kg for SMR-CC, and 5 L/kg for unabated SMR).

To support a total hydrogen output of 2.128 Mt (70.91 TWh), total fresh water demand is 15.96 Million m^3/year (Table 10). Despite this volume, desalination imposes a negligible economic and energetic burden on the system: annual desalination costs total 23.93 Million €/year, accounting for just 0.707% of total system expenditure and adding a marginal cost of only 0.0112 €/kg_{H_2}. From an energy perspective, the required 55.85 GWh of electricity consumption yields a specific energy overhead of 0.0262 kWh/kg_{H_2}, representing a mere 0.050% of overall system electricity demand. Consequently, while coastal desalination infrastructure is essen-

tial, water supply does not act as an economic or energetic bottleneck for hydrogen export.

Table 10: Dynamic multi-technology water consumption and desalination summary for the hybrid scenario.

Desalination & Water Consumption Metric	Hybrid Scenario
Total H_2 Output (TWh_{H2} / Million Tonnes)	70.91 / 2.128
Total Water Demand (Million m^3)	15.96
Estimated Desalination Cost (Million €/yr)	23.93
Desalination Cost Share of Total System (%)	0.707%
Desalination Marginal Cost (€/kg_{H2})	0.0112
Desalination Energy Overhead (GWh)	55.85
Desalination Specific Energy (kWh/kg_{H2})	0.0262
Desalination Share of Total Electricity Demand (%)	0.050%

Note: Based on an assumed seawater desalination cost of 1.50 €/m^3 and technology-specific water intensities.

7. Conclusion

PyPSA-Libya provides the first open-source, high-resolution look at Libya's energy future. The results suggest that Libya could become a competitive exporter of green hydrogen to Europe, provided that transmission bottlenecks are addressed.

Code and data availability

The model code is available at `https://github.com/mhdella/pypsa-libya`.

Appendix A. Optimization problem scale and computational complexity

The sector-coupled optimization model for Libya is formulated as a Linear Program (LP) using `linopy` with 2,920 time snapshots (8,760 hours at a 3-hour resolution) across 12 spatial clusters. Table A.1 summarizes the structural dimensions of the linear programming matrices for the three analyzed scenarios.

The primary driver of the dimensional variation across scenarios is the number of enabled hydrogen production technologies. In PyPSA, each active technology instantiated across network nodes generates both capacity investment variables and operational dispatch decision variables for every time step. Scenario A restricts

Table A.1: Linear programming matrix dimensions across the modeled scenarios.

Scenario	Decision Variables	Constraints	Non-Zero Entries
Scenario A	2,228,414	4,596,589	9,611,971
Scenario B	2,263,466	4,666,680	9,752,143
Scenario C	2,298,518	4,736,772	9,962,395

production exclusively to `H2 Electrolysis`, Scenario B includes `H2 Electrolysis` and `SMR CC`, and Scenario C permits all three options (`H2 Electrolysis`, `SMR`, and `SMR CC`). Excluding a technology skips component instantiation; each additional SMR technology adds exactly 35,052 decision variables ($2,920 \text{snapshots} \times 12 \text{nodes} +$ investment variables) and $70,092$ constraints to the matrix.

References

[1] M. Almaktar, M. Shaaban, Prospects of renewable energy as a non-rivalry energy alternative in libya, Renewable and Sustainable Energy Reviews 143 (2021) 110852. doi:10.1016/j.rser.2021.110852.

[2] K. Alanazi, S. Mittal, A. Hawkes, N. Shah, Renewable hydrogen trade in a global decarbonised energy system, International Journal of Hydrogen Energy 101 (2025) 712–730. doi:10.1016/j.ijhydene.2024.12.452.

[3] G. S. Eljrushi, T. N. Veziroglu, Solar-hydrogen energy system for libya, International Journal of Hydrogen Energy 15 (12) (1990) 885–894. doi:10.1016/0360-3199(90)90077-C.

[4] H. Abdel-Khalek, L. Schumm, E. Jalbout, M. Parzen, C. Schauß, D. Fioriti, Pypsa-earth sector-coupled: A global open-source multi-energy system model showcased for hydrogen applications in countries of the global south, Applied Energy 383 (2025) 125316. doi:10.1016/j.apenergy.2025.125316.

[5] A. Rahil, R. Gammon, N. Brown, Techno-economic assessment of dispatchable hydrogen production by multiple electrolysers in libya, Journal of Energy Storage 16 (2018) 46–60. doi:10.1016/j.est.2017.12.016.

[6] C. Gallego-Castillo, M. Victoria, Pypsa-spain: An extension of pypsa-eur to model the spanish energy system, Energy Strategy Reviews 60 (2025) 101764. doi:10.1016/j.esr.2025.101764.

[7] M. Parzen, H. Abdel-Khalek, E. Fedotova, M. Mahmood, M. M. Frysztacki, J. Hampp, L. Franken, L. Schumm, F. Neumann, D. Poli, et al., Pypsa-earth. a new global open energy system optimization model demonstrated in africa, Applied Energy 341 (2023) 121096.

[8] J. Fakhreddine, P. E. Dodds, I. Butnar, Global hydrogen trade pathways: A review of modelling advances and challenges, International Journal of Hydrogen Energy 129 (2025) 236–252. doi:10.1016/j.ijhydene.2025.04.203.

[9] A. Fattahi, F. D. Longa, B. van der Zwaan, Opportunities of hydrogen and ammonia trade between europe and mena, International Journal of Hydrogen Energy 83 (2024) 967–974. doi:10.1016/j.ijhydene.2024.08.021.

[10] N. Johnson, M. Liebreich, D. M. Kammen, P. Ekins, R. McKenna, I. Staffell, Realistic roles for hydrogen in the future energy transition, Nature Reviews Clean Technology (2025). doi:10.1038/s44359-025-00050-4.

[11] T. Terlouw, C. Bauer, R. McKenna, M. Mazzotti, Large-scale hydrogen production via water electrolysis: a techno-economic and environmental assessment, Energy & Environmental Science 15 (2022) 3583–3602. doi:10.1039/d2ee01023b.

[12] A. Achhammer, D. Fioriti, A. Patonia, M. Sterner, The impact of hydrogen underground storage on export dynamics and system resilience: A gis-based integration of salt caverns into pypsa-earth, Energy 360 (2026) 141729. doi:10.1016/j.energy.2026.141729.